********************* Translation in English of the French original ERB application *********************
********************************* Accepted on February 1st, 2016 *************************************
**************** Université du Québec à Trois-Rivières, Trois-Rivières, Québec, Canada *******************

1. Identification

Project title: Changes in spinal stiffness with chronic thoracic pain: correlation with pain and muscle activity

2. Project

2.1 Summary

The main goal of this project is to assess the reliability of thoracic spinal stiffness. To achieve this, 25 heartily participants (no significant thoracic pain in the past year) and 25 participants with chronic thoracic pain will participate in 2 experimental sessions separated by 24 to 48 hours. These sessions will occur at the Neuromecanic and Motor Control Laboratory of the Université du Québec à Trois-Rivières (Trois-Rivières, Québec, Canada). During these sessions, spinal stiffness will be measured at four thoracic spinal levels (T5 to T8) in a randomized sequence via a mechanical device using a servo-linear motor. The device indenter (metallic cylindrical cover with a rubber of 2 cm diameter) will gradually apply 45 N to the spinous process. The muscle activity and displacement data will be recorded during the procedures.

2.2 Objective, hypothesis or research question

The study proposed study is based on the research that has been conducted at UQTR during the past 5 years as part of our research group activities (Groupe de Recherche sur les Affections Neuromusculosquelettiques (GRAN)). The main goal of this project is to assess the reliability of thoracic spinal stiffness measure using the GRAN device. We believe that the measure will present a good within and between day reliability in both participants with and without chronic thoracic pain.

3. Methods

3.1 Experimental design

A controlled experimental study

3.2 Activities involving the participants

The participants will take part in two sessions (duration: 60 minutes) at the Neuromecanic and Motor Control laboratory of the UQTR. These sessions will be conducted at intervals of 24 to 48 hours. At the start of the first session, a short evaluation period including a physical examination and questionnaires (approximately 20 minutes) will allow participants eligibility confirmation. Participants will also complete the questionnaires at the beginning of the other session.

3.3 Research tools

For each tool used as part of your study, give a detailed description including: the justification of its utilization, its validity (if possible), the frequency of administration, where it will be done, and the time required to do it.

- A device using an actuator motor will allow the development and accurate delivery of force-time profiles aiming to assess spinal stiffness.
    - Spinal stiffness: This parameter will be measured at the spinous process of T5, T6, T7 and T8. The assessment consists of the application of a 45 N load. One procedure includes the assessment of each spinous process four times. Each measurement is of a duration of less than 10 seconds and a rest of 1 minute will be provided between each measurement.
- A system combining surface electromyography electrodes (sEMG) and accelerometers (Delsys) will be used to record the muscle response and the vertebral displacement during the assessment of spinal stiffness. To reduce skin impedance, the skin will be gently shaved, abraded with soft grade sand paper (Red Dot Trace Prep, 3 M, St. Paul, MN, USA), and clean with alcohol swabs. The electrodes will be positioned bilaterally over the erector spinae muscles at 2 cm of T5 and T8 spinous processes and at the level of T6/T7 interspinous space. Preparation duration: 5 minutes.
- Questionnaires :
    - The Quebec Back Pain Disability Scale (QBPDS), the Tampa Scale for Kinesiophobia (TSK), a visual analogue scale of 101 points (VAS), the STarT Back Screening Tool (SBT) and a pain body diagram will be completed by the participants. The QBPDS evaluates the impact of back pain on daily living activities [1]. The TSK evaluates fear of movement [2]. The VAS is a validated tool measuring pain level and is commonly used [3]. The pain body diagram will show the localization of the participant's symptoms. Expectations toward the intervention effect on the participant condition will also be assessed using a questionnaire. Questionnaires will be completed at the start of each session (duration: 10 minutes). An identification sheet will also be completed by the participants at the start of the first session to gather general data: age, sex, weight, height and thoracic back pain history.
    - Pain intensity will also be evaluated orally (0 to 100 scale) immediately following each assessment of spinal stiffness and each SMT.

3.4 Plan for data analysis

Describe the variables and the statistics methods that will be used:

The within and between day reliability of both spinal stiffness coefficients will be evaluated using intraclass correlation coefficients. The coefficient varies between 0 and 1 and the reliability is stronger as the value is near 1. Reliability will be determined for each group and at each spinal level. Statistical significance will be set at $p \leq 0.05$.

Correlations between the spinal stiffness (global and terminal coefficients) and both the muscle response amplitude (nRMS of the sEMG signals) and the pain intensity during the assessment of spinal stiffness will be calculated.

4. Participants

4.1 Number of participants required and sample size justification

The number of participants is based on the criteria presented by Shoukri et al. (2004).[4] Using the results of Wong et al. (2013) (ICC = 0.98), acceptable ICC value of 0.90, a statistical significance wet at 0.05 and a power of 0.80, a minimal of 11 participants per group was estimated. However, a conservative sample size was determined and set at 20 to 25 participants per group.

4.2 Inclusion criteria

Healthy participants: Adults aged between 18 and 60 years old not reporting any significant pain in the thoracic region in the past year or any thoracic pathology.

Participants with chronic thoracic pain: Adults aged between 18 and 60 years old presenting a constant or recurrent pain located between the first thoracic vertebra spinous process (T1) superiorly, the last thoracic vertebra spinous process (T12) inferiorly and laterally by the lateral border of the erector spinae muscles.[5] The pain has to be present since at least 12 weeks and of a non specific and non organic origin.

4.3 Exclusion criteria

Inflammatory spinal arthritis, abdominal or thoracic aorta aneurism, collagenases, advanced osteoporosis, spine surgery, neuromuscular disease, malignant tumor, uncontrolled hypertension, radiculopathy, progressive neurologic deficit, myelopathy, thoracic herniated disc, infection, thoracic scoliosis (Cobb's angle >20°), presence of any contraindication to a SMT of the thoracic spine and being a pregnant woman.

4.4 Recruitment strategy

Indicate how you will obtain participants contact information and how you will contact them :
Participants will be recruited within the university community (word-of-mouth strategy: research team). Social media (Facebook and UQTR En-Tête) will also be used.

If applicable, indicate where and by who will be done the tests or exams require to confirm participant eligibility:
A chiropractor of more than 5 years experience (Isabelle Pagé) will evaluate the participants at the start of the first session to confirm their eligibility.

Indicate the strategy that you will use to select participants and/or to create the groups (control/experimental):
Participants with chronic thoracic pain will first be recruited using convenience sampling strategies. Healthy participants will then be recruited by matching participants with chronic thoracic pain on their age and sex.

4.5 Study presentation

Indicate when, where and how the study will be presented to the participants:
The study will be presented during a phone interview with the participant and will be completed at the Neuromecanic and Motor Control Research Laboratory (3602-CH) at the start of the first session.

4.6 Compensation

Does the study include a compensation?
Yes

Please indicate the compensation and when it will be given to the participants.
25$ will be given to the participants at the end of the first session to thank them for their participation.

4.7 Consent of participants

Will you get a written consent to participate of the participants?
Yes

If yes, indicate how this consent will be acquired:
Participants will read the information sheet will be invited to ask any question related to the study and will then provide a written consent to participate. This consent will be provided at the research laboratory at the start of the first session.

If no, justify the impossibility to acquire consent:
Not applicable

References

Indicate the references associated with your research project (maximum of 10 references):

1. Kopec JA, Esdaile JM, Abrahamowicz M, Abenhaim L, Wood-Dauphinee S, Lamping DL, et al. The Quebec Back Pain Disability Scale. Measurement properties. Spine (Phila Pa 1976). 1995;20(3):341-52. PubMed PMID: 7732471.
2. Lundberg M, Grimby-Ekman A, Verbunt J, Simmonds MJ. Pain-related fear: a critical review of the related measures. Pain Res Treat. 2011;2011:494196. doi: 10.1155/2011/494196. PubMed PMID: 22191022; PubMed Central PMCID: PMCPMC3236324.
3. Ostelo RW, de Vet HC. Clinically important outcomes in low back pain. Best Pract Res Clin Rheumatol. 2005;19(4):593-607. doi: 10.1016/j.berh.2005.03.003. PubMed PMID: 15949778.
4. Shoukri MM, Asyali MH, Donner A. Sample size requirements for the design of reliability study: review and new results. Stat Methods Med Res. 2004;13(4):251-71. doi: 10.1191/0962280204sm365ra. PubMed PMID: WOS:000223201100001.
5. Merskey H, Bogduk N, International Association for the Study of Pain. Task Force on Taxonomy. Classification of chronic pain : descriptions of chronic pain syndromes and definitions of pain terms. 2nd ed. Seattle: IASP Press; 1994. xvi, 222 p. p.

**************************** Amendment to the initial project request ******************************

************************************** May 11th, 2017**

For each amendment, clearly defined the situation described in the initial application, the medication you would like to make and the justification of this modification.

In the initial application, the project included 3 groups of 25 participants with chronic thoracic pain. We would like to add a fourth group of 25 participants with chronic thoracic pain. This group will control for the cyclic nature of chronic back pain. Moreover, it will control for the possibility of a clinical change associated with the participation to the study or with the light pressures over the back during the assessment of spinal stiffness. These participants will take part in the same number of sessions (i.e. 4 sessions) and these sessions will be separate as the sessions of the other groups. However, during the first three sessions, these participants won't receive a spinal manipulative therapy. Spinal stiffness will first be assessed, followed by a 5 minutes rest and, finally, spinal stiffness will be assessed de novo. The fourth session will be identical to the fourth session of the other groups.

What are the consequences of these amendments on the risks faced by the participants and what means have been deployed to minimized them?

There is no modification to the risks faced. Indeed, the participants of this group might also feel minor side effects such as an increased pain that should resolved within 48 hours. The time dedicated to the project, around 60 minutes per session, remains a disadvantage.

Do the participants already recruited for the study need to be contacted? If so, how the contact will be made?

The recruitment is not yet started. It will start before the end of May.